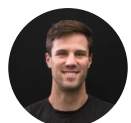


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Estimate your 1RM with bar speed data in Metric

Estimate your 1RM with Metric VBT. Use velocity based training (VBT) to calculate an e1RM every workout



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Automatically estimate 1RM from barbell velocity with the Metric VBT app

For technical details about how to use the estimated 1RM feature in the



Metric VBT app check out this support guide →

Your 1RM represents the heaviest weight you can lift for a single full repetition with proper form for a given exercise. It is the ultimate measure of your strength on an exercise and an important motivator for lifters whether they compete in powerlifting, bodybuilding or simply love training with weights!

However, testing your true 1RM can be a time consuming, taxing and even risky process. Which is where the estimated 1RM (e1RM) comes in, a substitute value that can be derived from your barbell velocity data without having to perform a maximum set. It is easy to calculate your estimated 1RM during a normal workout tracking barbell velocity on your warmup sets using the Metric VBT app.

Metric will then automatically and accurately provide your exercise-specific estimated 1RM using velocity-based training (VBT) principles providing you with real-time insights into your maximum strength levels, fatigue and progress.

Calculate
your
estimated
1RM
automatically
in the Metric
VBT app.
Using just the
bar speed
data
collected on
your phone

What is an estimated 1RM

Knowing how much weight you can lift for your one repetition maximum (also written as 1RM, 1 rep max, and one rep max) is a powerful piece of information for programming and tracking progress in your strength training workouts.

An estimated 1RM (e1RM) is the fast and easy way to find your strength levels during a workout, without having to actually do a 1RM.

Collecting an e1RM score requires no additional reps or sets outside your normal training session. Simply track velocity (In Metric we use mean velocity in m/s for this) with Metric VBT across three or more sets on the given exercise and the app will provide a predicted 1RM value without the need for maximal testing, making it a safer and more convenient option to calculating your maximum strength levels in real time..

How to calculate your 1RM with VBT

This section is a technical look at the maths that goes into calculating an estimated 1RM from bar speed data. You don't have to do anything with this information as the Metric app estimates your 1RM automatically as you train. This might just be useful information to know!

The Load-Velocity Profile

To calculate your 1RM with velocity based training (VBT) requires the creation of a load-velocity profile.

When performing strength exercises, as the load increases, the velocity

of the bar decreases. This inverse relationship between load and velocity is called the load-velocity profile. In Metric VBT your load velocity profile is generated automatically during your workout, showing your own unique profile for that exercise and comparing it to your profile from last workout at the same time.

A load
velocity
profile
collected
from
barbell
tracking
data in
the
Metric
VBT app

This relationship between load and velocity is stable, linear, and repeatable for most traditional barbell exercises (the exception being the Olympic Weightlifting movements). Because of this stability and predictability it becomes a great tool for measuring training performance in the gym.

For your load velocity profile to be calculated and shown in the app, you just need three or more sets across a range of loads for a single exercise in a workout. A typical warmup series of sets is perfect for creating your profile.

Fast reps – Lifting with intent

For your profile to be truly reflective of your performance and to get an

accurate e1RM in the next steps it's important that you apply high levels of intent to these reps, trying to lift the weight as fast as possible while maintaining good form for an accurate gauge of your strength and power levels.

These reps don't have to be so fast they become dangerous (especially for weights under 50% of 1RM) but you need to be giving some effort – trying to lift at a pretty decent level of intent on all weights to give you a profile that will be useful for estimating your 1RM.

Velocity at your 1RM – the minimum velocity threshold (MVT)

To find your e1RM, we need to establish a load-velocity profile and extend the linear trend line out to the point where it reaches 100% of your 1RM.

But because we are calculating the 100% point on the trend line with this process, we need one more variable, the bar velocity at this 100% point. The velocity at this 100% of 1RM point is called the minimum velocity threshold (MVT), it is the slowest speed at which you can still actually move a weight and complete the rep. Any heavier (and therefore any slower) you would fail the rep.\

The
estimated
1RM
intersects
with the
velocity
at your
point of
failure,

also
known as
the
minimum
velocity
threshold
(MVT)

With your minimum velocity threshold (MVT), the load velocity profile, and a good amount of intent applied along the way, you can now accurately pinpoint the load that corresponds with 100% of your 1RM.

 You have now estimated your 1RM without actually attempting a maximal lift.

While it might seem like there is a lot of maths and calculations required to find your e1RM, this entire process is automated by the Metric VBT app. All of the calculations and profiling is handled by the app, taking your training data and providing an e1RM value directly in the workout page. The app then updates this value with every set completed, or if you change the MVT value in your exercise settings.

Selecting a barbell velocity for your 1RM

While the 1RM calculation is automatic, the part you will have to handle manually is knowing what mean velocity value you should use as your MVT for each exercise.

While finding the exact MVT value for all of your lifts might seem desirable, it's important to understand that daily fluctuations in readiness, fatigue, and motivation can all affect your MVT. Rather than chasing an elusive "perfect" MVT, you are better off picking a value that

is in the right range and sticking with this one value for that exercise as a consistent reference point to estimate your daily strength. Most lifters should be able to find an MVT with a high degree of accuracy, typically within 1-2% each session.

Velocity based training chart for finding your velocity at 1RM

The table below provides a guide to finding the minimum velocity threshold velocities for your favourite exercises. Your velocity might vary slightly from these values depending on factors such as height, range of motion, technique, lifting experience, and equipment.

Here's a table with a range of velocities at 1RM for various exercises:

Exercise	Novice	Intermediate	Advanced
Back Squat	0.4	0.3	0.25
Barbell Row	0.5	0.4	0.3
Bench Press	0.3	0.2	0.12
Deadlift	0.35	0.2	0.15
Front Squat	0.4	0.35	0.3
Overhead Press	0.35	0.25	0.15
Sumo deadlift	0.3	0.2	0.1
Trapbar deadlift	0.45	0.35	0.3

Unless you are a powerlifter using this MVT to make competition attempt selections, an MVT value that is close enough should be more than sufficient. I would advise against changing your MVT value more than once per year.

Calculate Your 1RM with the Metric VBT app

Once you have determined the appropriate MVT for your exercises, the Metric VBT app makes it easy to estimate your 1RM by analysing your load-velocity profile and identifying the weight corresponding to your MVT.

Using Metric VBT's 1RM estimation feature is simple:

1. **Set your 1RM velocity value in the exercises settings** of this is accessible from an active workout by pressing the menu button beside the exercise name. Toggle on e1RM and enter the velocity value you want to use for this exercise. the MVT is in mean velocity and refers to the speed on the concentric (up) phase of each rep.
2. **Perform at least 3 sets** of the exercise across a range of loads and velocities. Record barbell velocity for each set using Metric VBT.
3. **Lift with intent** on all sets and rest sufficiently between sets for accurate velocity measurements.
4. **The app will generate your load-velocity profile** and estimate your 1RM once enough sets have been recorded. Metric uses the best rep velocity from each set to calculate your load-velocity profile and e1RM.
5. **Track progress on the e1RM over time** on the Performance tab within Metric, you can review how your e1RM has changed over time, assessing if your training is heading in the right direction or making adjustments to your program as needed.

Benefits of 1RM Estimation

Estimating your 1RM using Metric VBT offers numerous advantages over traditional 1RM testing.

By utilising submaximal loads and velocity data, you can safely and efficiently assess your strength during your regular training sessions without the risks, fatigue, and inconvenience associated with maximal attempts.

This approach allows you to track progress more frequently, modify your training plan with up-to-date information, make real-time program adjustments, and ensures that you're always working at the appropriate intensity for your goals.

Improve your percentage-based training with VBT

Knowing your e1RM enables you to prescribe training loads as a percentage of your maximum with real-time 1RM data, allowing for more precise programming and progression.

Using velocity-based training to find an e1RM in real-time ensures you're working with the most up-to-date information, allowing you to calibrate your rate of progression and daily load prescription to account for daily fluctuations in strength, recovery rate and fatigue levels.

Utilising daily estimated 1RM values in your % based programming offers a significant advancement over traditional methods. Traditional 1RM testing, done infrequently, fails to account for the day-to-day fluctuations in an athlete's strength and readiness, which can lead to misjudged training intensities, risking stagnation for some athletes due to progression that is too slow, or overtraining and burnout for others that over index on testing day but can't keep up with the program in regular training.

Track progress without maxing out

Regularly estimating your 1RM with barbell velocity provides all the benefits of 1RM tests but without the drawbacks of actual sets to failure.

This saves weight room time by eliminating the need for separate testing sessions and helps maintain a consistent level of engagement and focus throughout your training cycle. Instead of starting each training block with a max out workout, you can instead get straight into the next training cycle and put in a productive training session.

This doesn't mean we have to completely forget about the hype and energy of a testing day.

By collecting an e1RM value every single workout it turns each session into a low-stakes testing event — allowing you to capture some of that testing day energy and bring that into your daily training. A little competition on e1RM (maybe made relative to bodyweight for a level playing field) will drive intent into your lifts, create banter in your gym, and keep everyone focused on putting in great, consistent training .

As you get stronger, your load-velocity profile will shift up and to the right, leading to an increase in your e1RM. In the Metric app, you can even watch this change in your profile with the last-workout comparison line. Serving as a tangible reminder of your gains, reinforcing your commitment to the training process.

Today's
load
velocity
profile
(yellow)
contrasted

with the
last time
profile line
in grey.
The goal is
to move
your
profile up
and to the
right as
you get
stronger
and more
explosive.

Monitoring Fatigue and Readiness with e1RM

Because the e1RM is derived from your load-velocity profile, it is sensitive to changes in the relationship between the weight lifted and the speed at which it moves. Fluctuations in your e1RM therefore serve as an accurate and objective early warning system for accumulated fatigue or impending plateaus in athletes who are not recovering and adapting to their program.

If your e1RM is stagnate or even declining despite consistent training, it may indicate that your body is struggling to recover between sessions. In this case, a program adjustment, more deliberate recovery work (get some sleep!) or a strategic deload and programming adjustment could help restore your strength and prevent overtraining.

On the other hand, if your e1RM is steadily increasing, it can give you the confidence to push hard in your training!

How accurate are estimated 1RM scores?

When a good spread of weights are used, lifts are performed with intent, and an appropriate MVT is used, your e1RM should be a reliable indicator of your strength levels on any given day. Numerous studies have demonstrated a strong correlation between VBT-based 1RM estimates and true 1RM values obtained from maximal testing.

A study by Jidovtseff et al. (2011) found that VBT-based 1RM estimates were within 5% of true 1RM values for the bench press, while a similar study by Banyard et al. (2018) reported an average error of less than 3% for the back squat.

While e1RMs may not match the absolute accuracy of a true 1RM test, the ability to test maximum strength frequently, safely, and efficiently and on a weekly basis without disrupting your program makes the e1RM a valuable tool for strength athletes and coaches.

Calculate your estimated 1RM with Metric VBT

Metric VBT requires no additional devices to track barbell velocity — just your smartphone.

Estimate your 1RM with remarkable accuracy, track progress effortlessly, and fine-tune your programming based on real-time data with Metric's high precision computer vision system.

Whether you're a powerlifter, an S&C coach, a competitive athlete, or a fitness enthusiast looking to take your strength to the next level, Metric VBT's e1RM feature is an invaluable tool for unlocking your full potential.

Metric VBT's e1RM feature enables more responsive programming, better fatigue management, and ultimately, faster progress toward

your strength goals.

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By lifters for lifters

Metric was built with lifters and coaches front of mind.

Focused analytics and powerful workout tracking tools for Powerlifting, Strength and conditioning, CrossFit, Weightlifting, and anyone serious about their strength training.



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